

ANDREW LIN

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EDUCATION

University of Waterloo

Sept. 2020 – 2025

Candidate for Bachelor of Computer Science

Waterloo, Ontario

- Relevant Courses: Operating Systems, Algorithms, AI, Distributed Systems, Concurrency and Parallelism

TECHNICAL SKILLS

Languages: C, C++, Python, Golang, JavaScript, C#.NET, Java

Technologies: Git, MongoDB, PostgreSQL, AWS, Grafana, Kubernetes, Docker, Prometheus, Ansible, Jenkins

Frameworks/Libraries: React, Node.js, Express.js, Socket.io, Django, MFC

WORK EXPERIENCE

Arctic Wolf Networks

July 2025 – Present

Software Developer

Waterloo, Ontario

- Developed an end-to-end observability pipeline using Vector and Prometheus to track provisioning production times across 20+ sensor variants identifying critical production bottlenecks
- Automated the production pipeline by codifying manual registration steps using Python into a centralized database and implementing Go-based services to dynamically sync with AWS reducing production deployments by 30%
- Modernized core provisioning infrastructure by orchestrating migration from Ubuntu 16.04 to 24.04, ensuring compatibility for Dockerized certificate services and secure VPN connectivity for 20000+ sensors

Arctic Wolf Networks

Sept. 2024 – Dev 2024

Software Developer Intern

Waterloo, Ontario

- Enabled a seamless migration to Karpenter across 6+ services by optimizing Kubernetes configurations and implementing signal handlers in Golang, resulting in a seamless migration and improving service reliability
- Automated PR creation for api-spec-driven code updates using Github Actions and shell scripts, improving productivity for 2+ projects by minimizing developer intervention by 50%
- Developed a Grafana visualization to track sensors and certificate expirations, enabling visibility of upcoming expirations for 20000+ sensors

Amazon (AWS)

May 2024 – Aug. 2024

Software Development Engineer Intern

Toronto, Ontario

- Implemented a Grafana dashboard for data visualizations on CPU and sudo usage, aiding in CPU optimization efforts and potentially saving the organization 2500 CPU hours of processing time across production
- Designed an automated system using AWS Lambda, AWS Athena, EventBridge, and S3 to orchestrate CPU profiling for the dashboard, leading a 3x increase in time efficiency
- Wrote bash scripts to access CPU usage, facilitating data collection and analysis for performance improvements, leading to a 70% increase in CPU data

Geotab

May 2023 – Aug. 2023

Software Developer Intern

Oakville, Ontario

- Integrated a new vehicle database in C#.NET and Entity Framework into Geotab's data ingestion service by writing functions to add, update, and delete from a CloudSQL database, resulting in over 50000+ added vehicles
- Modified data ingestion service to utilize new Google Protobuf message to send data to Google Pub/Sub, resulting in over 140M+ messages a day serializing and deserializing to the updated Google Protobuf message
- Refactored code to utilize Dependency Injection using C#.NET across vehicle data ingestion data flow to help facilitate the migration of the application from the VM environment to Kubernetes

Rocscience

Sept. 2022 – Dec. 2022

Software Developer Intern

Toronto, Ontario

- Utilized C++, MFC, and Codejock to overhaul their 3D CAD software's user interface, by implementing an updated ribbon bar, dropdown buttons, and an updated combobox, contributing to a successful product release
- Converted desktop application, file, and registry classes into NuGet Packages in C++, improving code maintainability and saving 10% of development time

PROJECTS

My-Chat-App | React, Node.js, Express.js, Socket.io, MongoDB

- Built a web-based chat application using the MERN stack that allows users to chat with other users privately

ASCII Game Engine | C++, Ncurses

- Implemented a C++ game engine to support the creation of ASCII art video games